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Week 13 - Final Project Phase 4 | Reflection on Emerging Database Technologies

In modern database systems, new technologies are improving how data is stored, processed, and managed. Two emerging trends that are very useful today are **distributed databases** and **machine learning (ML)-integrated databases**. These technologies can help improve the performance and reliability of our current group project.

Distributed databases store data across multiple servers or locations instead of only one system. This helps improve speed and availability because if one server fails, another can still provide the data. For our project, this means better reliability and faster access, especially if many users are using the system at the same time. It also helps when handling larger amounts of data since the workload is shared across servers.

ML-integrated databases use machine learning to improve how data is managed. For example, they can automatically optimize queries, detect unusual activity, and even predict future data trends. In our project, this could help make the system smarter by improving search speed and helping analyze data more efficiently without manual work.

However, there are also limitations when using these technologies. Distributed databases can be difficult to set up and manage because data must stay synchronized across all servers. If not properly configured, it can lead to data inconsistency. It also requires more technical knowledge, which can be challenging for beginners.

For ML-integrated databases, one limitation is the need for large amounts of data to work effectively. Without enough data, the machine learning models may not produce accurate results. It can also increase system complexity and require more computing resources, which may not always be available in free cloud services.

Overall, these emerging technologies can greatly improve our group project by making it faster, smarter, and more reliable. However, they also introduce challenges in terms of setup, cost, and complexity. Understanding both the advantages and limitations helps us design better and more realistic database systems for future use.